Yalan Song (she/her)

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Academic Experience

2023-present	Research Assistant Professor	Pennsylvania State University
2022-2023	Postdoctoral Fellow	Pennsylvania State University

Education

2017-2022	PhD	Pennsylvania State University	Water Resources Engineering (Major)
2017-2022	PhD	Pennsylvania State University	Computational Science (Minor)
2014-2017	MS	Shanghai Jiao Tong University	Ocean Engineering
2010-2014	BS	Tianjin University	Ocean Engineering

Funding Experience

- **External Funding**
  - **Funded**
    - *Surface-subsurface Integrated Water Prediction System for Long-Term Assessment with Physics-Informed Differentiable Modeling*, funded by USGS (2024-2026), Role: Project co-PI, $640,000
    - *Arctic Hydrologic and Thermal Dynamic Modeling*, LANL subaward (2025-2029), Role: Project co-PI, $600,000
    - *Differentiable Hybrid Hydrologic Modeling for RFC Forecasting Operations*; invited for full proposal submission to NOAA in 2026; Role: PSU lead PI, $300,000 (2026 July-2027 March)
    - *CIROH Scoping Project: Large-domain Hydrologic Model Calibration Initiative*; invited for full proposal submission to NOAA in 2026; Role: Project co-PI, $300,000 (2026 July-2027 March)
  - **Pending**
    - *From Coastal Flooding to Invisible Salinity Risk: A NASA-Enabled Decision-Support Framework for Coastal Farmland Protection*; Submitted to NASA in 2026; Role: Project co-PI
    - *A Differentiable Framework for Regional Surface-Subsurface Reactive Transport*; Submitted to USGS in 2026; Role: Project lead PI
    - *Foundation AI with differentiable surface-subsurface integrated model for model-informed monitoring-network design at the national scale*; Submitted to USGS in 2026; Role: Project co-PI
    - *An AI-Ready Compendium for Flash Flood Modeling Experiments and Benchmarking in the NextGen Framework*; Submitted to NOAA in 2026; Role: PSU lead PI
    - *Compound Low-Flow and Thermal Stress Risks to Energy Systems and Data Centers*; Submitted to NOAA in 2026; Role: PSU lead PI
    - *Developing Practical Strategies for Differentiable Hydrologic Model Training in the RFC Context*; Submitted to NOAA in 2026; Role: PSU lead PI
    - *Probabilistic ensemble forecasting and data assimilation with differentiable models*; Submitted to NOAA in 2026; Role: Project co-PI

Research Experience

- **Data-driven and hybrid Models in large-scale hydrology**

  - The AI-based high-resolution differentiable (physics-informed) hydrologic modeling framework for the next generation of Operational National Water Model.
  - Differentiable modeling of coupled surface and subsurface hydrologic processes for integrated water prediction.
  - An AI-based differentiable hydrologic model for improving the prediction of unseen extreme flood events.
  - The adjoint-based implicit solver to enable advanced numerical schemes in the large-scale differentiable hydrologic modeling.
  - Differentiable groundwater modeling for high-dimensional parameter inversion
  - Long Short-Term Memory (LSTM) models in hydrology and environmental science for regional snow water equivalent and CONUS-scale suspended sediment concentration.
- **Data-driven and hybrid models in computational hydraulics**
  - AI-based surrogate models for shallow water equations (SWEs) in computational hydraulics.
  - River channel bathymetry inversion using an AI-based surrogate model of SWEs.
  - AI-based differentiable SWEs for learning the distribution of key hydraulic parameters (roughness) in the river channel.
- **Classic computational environmental fluid dynamics**
  - 3D local scour (hydrodynamic+sediment transport) solver, *ibScourFoam*, for scour modeling around complex structures using OpenFOAM (C++).
  - 3D scour model for the remediation of underwater unexploded ordnances
  - Review paper about 3D scour modeling
  - Physically-based sand slide method for scour models.
- **Mentored research**
  - Variational data assimilation for AI-based differentiable hydrologic models to improve streamflow forecasts.
  - Using AI-based high-resolution differentiable hydrologic modeling framework for global water assessment.
  - An ensemble of differentiable hydrologic models for a multi-model mosaic framework.

Publications

- First authorship papers
  - **Song, Y.**, Sawadekar, K., Frame, J.M., Pan, M., Clark, M.P., Knoben, W.J., Wood, A.W., Lawson, K.E., Patel, T. and Shen, C., 2026. Physics-informed, Differentiable hydrologic models for capturing unseen extreme events. Water Resources Research, 62(2), p.e2025WR040414. https://doi.org/10.1029/2025WR040414
  - **Song, Y.**, Bindas, T., Shen, C., Ji, H.*, Knoben, W. J. M., Lonzarich, L.*, et al., 2025. High-resolution national-scale water modeling is enhanced by multiscale differentiable physics-informed machine learning. Water Resources Research, 61, e2024WR038928. https://doi.org/10.1029/2024WR038928
  - **Song, Y.**, Knoben, W.J., Clark, M.P., Feng, D., Lawson, K., Sawadekar, K.* and Shen, C., 2024. When ancient numerical demons meet physics-informed machine learning: adjoint-based gradients for implicit differentiable

modeling. Hydrology and Earth System Sciences, 28(13), pp.3051-3077, https://doi.org/10.5194/hess-28-3051-2024.

  - **Song, Y.**, Chaemchuen, P., Rahmani, F., Zhi, W., Li, L., Liu, X., Boyer, E., Bindas, T., Lawson, K. and Shen, C., 2024. Deep learning insights into suspended sediment concentrations across the conterminous United States: Strengths and limitations. Journal of Hydrology, p.131573, https://doi.org/10.1016/j.jhydrol.2024.131573
  - **Song, Y.**, Tsai, W.P., Gluck, J., Rhoades, A., Zarzycki, C., McCrary, R., Lawson, K. and Shen, C., 2024. LSTM-based data integration to improve snow water equivalent prediction and diagnose error sources. Journal of Hydrometeorology, 25(1), pp.223-237, https://doi.org/10.1175/JHM-D-22-0220.1.
  - **Song, Y.** , Shen, C. and Liu, X., 2023. A surrogate model for shallow water equations solvers with deep learning. Journal of Hydraulic Engineering, 149(11), p.04023045, https://doi.org/10.1061/JHEND8.HYENG-13190.
  - **Song, Y.**, Xu, Y., Ismail, H. and Liu, X., 2022. Scour modeling based on immersed boundary method: A pathway to practical use of three-dimensional scour models. Coastal Engineering, 171, p.104037, https://doi.org/10.1016/j.coastaleng.2021.104037.
  - **Song, Y.**, Darzikolaei, S.A.M. and Liu, X., 2022. Scour around underwater unexploded ordnances (UXOs): An experimental and computational investigation. Ocean Engineering, 262, p.112146, https://doi.org/10.1016/j.oceaneng.2022.112146.
  - **Song, Y.**, Xu, Y., & Liu, X. (2020). Physically Based Sand Slide Method in Scour Models Based on Slope-Limited Diffusion. Journal of Hydraulic Engineering, 146(11), 04020074, https://doi.org/10.1061/(ASCE)HY.1943-7900.0001814.
  - **Song, Y.**, Lai, Y.G. and Liu, X., 2020. An improved immersed boundary method for simulating flow hydrodynamics in streams with complex terrains. Water, 12(8), p.2226. https://doi.org/10.3390/w12082226

- Mentees (noted with *)' papers
  - Ji, H.*, **Song, Y.**, Bindas, T., Shen, C., Yang, Y., Pan, M., Liu, J., Rahmani, F., Abbas, A., Beck, H. and Lawson, K., 2025. Distinct hydrologic response patterns and trends worldwide revealed by physics-embedded learning. Nature Communications, 16(1), p.9169. https://www.nature.com/articles/s41467-025-64367-1
  - Jamaat, A.*,**Song, Y., corresponding author**, Rahmani, F., Liu, J., Lawson, K. and Shen, C., 2025. Update hydrological states or meteorological forcings? Comparing data assimilation methods for differentiable hydrologic models. Journal of Hydrology. https://doi.org/10.1016/j.jhydrol.2025.134137
  - Sawadekar, K.*, **Song, Y. corresponding author**, Pan, M., Beck, H., McCrary, R., Ullrich, P., Lawson, K. and Shen, C., 2025. Improving differentiable hydrologic modeling with interpretable forcing fusion. Journal of Hydrology, p.133320. https://doi.org/10.1016/j.jhydrol.2025.133320

- Collaborative papers
  - Clark, M.P., Knoben, W.J., Spieler, D., Grundemann, G.J., Thebault, C., Vasquez, N.A., Wood, A.W., **Song, Y.**, Shen, C., Carney, S. and van Werkhoven, K., 2026. Comment on: "Friends don't let friends use NSE or KGE for hydrologic model accuracy evaluation: A rant with data and suggestions for better practice". Environmental Modelling & Software, p.106869.

https://doi.org/10.1016/j.envsoft.2026.106869

- Li, P., **Song, Y.**, Pan, M., Lawson, K. and Shen, C., 2025. Ensembling differentiable process-based and data-driven models with diverse meteorological forcing datasets to advance streamflow simulation. Hydrology and Earth System Sciences, 29(23), pp.6829-6861. https://doi.org/10.5194/hess-29-6829-2025
- Liu, X. and **Song, Y. corresponding author**, 2025. Scientific machine learning of flow resistance using universal shallow water equations with differentiable programming. Water Resources Research, 61(9), e2025WR040265. https://doi.org/10.1029/2025WR040265
- Knoben, W.J., Raman, A., Grundemann, G.J., Kumar, M., Pietroniro, A., Shen, C., **Song, Y.**, Thebault, C., van Werkhoven, K., Wood, A.W. and Clark, M.P., 2025. How many models do we need to simulate hydrologic processes across large geographical domains?. Hydrology and Earth System Sciences, 29(11), pp.2361-2375. https://doi.org/10.5194/hess-29-2361-2025
- Yang, Y., Pan, M., Feng, D., Xiao, M., Dixon, T., Hartman, R., Shen, C., **Song, Y.**, Sengupta, A., Delle Monache, L. and Ralph, F.M., 2025. Improving streamflow simulation through machine learning-powered data integration and its potential for forecasting in the Western US. Hydrology and Earth System Sciences, 29(20), pp.5453-5476. https://doi.org/10.5194/hess-29-5453-2025
- Liu, J., Shen, C., O'Donncha, F., **Song, Y.**, Zhi, W., Beck, H.E., Bindas, T., Kraabel, N. and Lawson, K., 2025. From RNNs to Transformers: benchmarking deep learning architectures for hydrologic prediction. EGUsphere, 2025, pp.1-21. https://doi.org/10.5194/egusphere-2025-1706
- Liu, X., **Song, Y.** and Shen, C., 2024. Bathymetry inversion using a deep-learning-based surrogate for shallow water equations solvers. Water Resources Research, 60(3), p.e2023WR035890. https://doi.org/10.1029/2023WR035890
- Shen, C., Appling, A. P., Gentine, P., Bandai, T., Gupta, H., Tartakovsky, A., ... & Lawson, K. 2023. Differentiable modelling to unify machine learning and physical models for geosciences. Nature Reviews Earth & Environment, 4(8), 552-567. https://www.nature.com/articles/s43017-023-00450-9
- Lai, Y.G., Liu, X., Bombardelli, F.A. and **Song, Y.**, 2022. Three-dimensional numerical modeling of local scour: A state-of-the-art review and perspective. Journal of Hydraulic Engineering, 148(11), p.03122002. https://doi.org/10.1061/(ASCE)HY.1943-7900.0002019

Preprints

- Shen, C. and **Song, Y.**, 2026. A Differentiable, Mass-Conservative Model of Coupled Unsaturated–Saturated Flow: Benchmark and High-dimensional Inversion. SSRN preprint, https://doi.org/10.2139/ssrn.7524427.
- Yang, W., Lonzarich, L., **Song, Y.**, Ji, H., Pan, M., Lawson, K., and Shen, C., 2026. Hourly U.S.-wide flood simulation beyond the limits of traditional and data-driven models. arXiv preprint, https://doi.org/10.48550/arXiv.2609.06794.
- **Song, Y.**, Clark, M., Liu, J., Halgren, J., Lawson, K., and Shen, C. 2024. Prominent impacts of hydrologic scaling laws on climate risks. Preprinted and submitted to Nature Water, https://doi.org/10.21203/rs.3.rs-4584048/v1.

Teaching Experience

- Teaching assistant for Computing Methods and Water Resources Engineering.
- Guest instructor of AI for Geosciences (grad level) at Penn State and AI for Water (grad level) at Tsinghua University
- Instructor of the CIROH Developer Conference (University of Utah), the Knowledge-Guided Machine Learning (KGML) Workshop (University of Michigan), and an online Deep Learning for Water workshop

Mentoring Experience

Mantee	Position	Publication	Other achievements
Kamlesh Sawadekar	Ph.D candidate	Journal of Hydrology	First place of poster at the 38th AMS
Amirmoez Jamaat	Ph.D candidate	Journal of Hydrology	-
Leo Lonzarich	Ph.D candidate	-	Outstanding Student Presentation at AGU 2024
Haoyu Ji	Ph.D candidate	Nature Communications	

Service

- The mentoring committee in Department of Civil and Environmental Engineering at Penn State.
- Primary convener and co-convener of two sessions in AGU 2025 Annual Meeting.
- Reviewers of Water Resources Research, Hydrology and Earth System Sciences, AGU advances, Engineering Applications of Computational Fluid Mechanics, Advances in Water Resources, Journal of Geophysical Research - Earth Surface, Journal of Hydrology, Journal of Hydraulic Engineering, The Cryosphere, etc
- NSE proposal reviewer.

Invited presentations

- Improving Operational Extreme Flood Forecasting and Climate Change Impact Assessment with Physics-Embedded Differentiable Modeling. EGU General Assembly 2026.
- Differentiable Physics-Informed Machine Learning Enhances High-resolution Hydrologic Modeling, AGU Annual Meeting 2025.
- High-resolution Differentiable Models for Operational National and Global Water Modeling Assessment, Plenary Keynote, CSDMS 2025 Annual Meeting.
- High-Resolution Differentiable Models for Operational National and Global Water Modeling and Assessment. EGU General Assembly 2025.
- Differentiable Modelling to Unify Machine Learning and Physical Models for Geosciences, AGU Chapman Remote Sensing of the Water Cycle: Sensors to Science to Society, Honolulu, HI, 2024.

Awards and Fellowships

2017	Outstanding graduates of Shanghai Jiaotong University, China
2014	American Bureau of Shipping Scholarship
2012	National Scholarship awarded by The Ministry of Education, China
2011-13	Honor Student of Tianjin University, China